

Lab Tasks STL and Hash Tables

Note: Perform the tasks 1-4 using STLs

Task 1:

A customer support center processes incoming support tickets. Tickets are handled in FIFO order, but urgent tickets may be inserted at the front.

Each ticket stores information about ticketID, customerName, issueType and priorityLevel (1 = normal, 2 = urgent)

Develop a system that:

- Adds new tickets:
- Serves the next ticket (removal operation)
- Displays all pending tickets
- Shows the current ticket being processed
- Counts how many tickets belong to a specific issueType
- Searches for a ticket by ticketID

Task 2:

A programming tool must validate whether an expression contains properly balanced parentheses. Input is a string expression containing brackets: (), {}, []

Implement a system that:

- Processes the expression using a stack
- Validates matching and ordering of brackets
- Reports whether the expression is balanced or unbalanced
- Additionally:
 - Count total number of valid bracket pairs
 - Detect first mismatch position (if any)

Task 3:

A retail system maintains a collection of products and needs analytical operations for reporting.

Each Product has productID, name, price and category.

Implement a program that:

- Stores products dynamically
- Sorts products by price using sort()

- Searches for a product by productID
- Counts how many products belong to a given category
- Applies a price increase using transform()
- Displays all products using for_each()

Task 4:

Two departments maintain independently sorted record datasets that must be merged.

Define a class Record with:

- int recordID
- string name
- double value

Develop a system that:

- Maintains two sorted collections of records based on value
- Merges them into a third collection using merge()
- Displays merged result
- Verifies that final data remains sorted
- Identifies duplicate recordID entries

Hash Tables:

Task 5:

Implement ADT of Hash Table with following functions: insert, delete, search and a hash function.

Task 6:

Create a Website login system. System should provide an interactive panel to register the users. User can login the system only if correct password is entered. Insert passwords into the Hash Table at the time of user registration. For login, retrieve the user's password from the Hash Table and compare retrieved user password to input password and print "Authentication failure" or "Authentication successful". Reuse the already created hash table of Task 5.